

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**  
**End Semester Examination – Winter 2022**

Course : B. Pharmacy  
Subject Name : Instrumental Methods of Analysis  
Max Marks : 75

Date: 03/01/2023  
Sem: VII  
Subject Code: BP701T  
Duration : 3 Hr.

**Instructions:**

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

**Q. 1. Answer following questions. (10 x 2) = 20**

- i) Write the electronic transitions for the following molecules.  
a) 1-3 butadiene b) acetaldehyde
- ii) Differentiate between singlet and triplet state.
- iii) Differentiate between single component and multicomponent analysis.
- iv) The compound A had travelled distance 5 cm, compound B distance travelled 7 cm on TLC plate. The solvent front distance was 10 cm. Calculate the R<sub>f</sub> value for compound A and B?
- v) What is quenching. Enlist its types.
- vi) Draw a diagram of single beam and double beam UV-Visible spectrophotometer.
- vii) Differentiate between normal phase and reversed phase chromatography.
- viii) Define with example: auxochrome, chromophore.
- ix) Why derivatization techniques required in gas chromatography? Enlist derivatization methods used in GC.
- x) Calculate the concentration of compound in an ethanolic solution of which the absorbance in a 1cm cell at its  $\lambda_{\text{max}}$  241 nm, was found to be 0.890. The A (1%, 1cm) of compound D is 540 at 241nm.

**Q. 2. Answer the following questions (any two) (2 x 10) = 20**

- i) Explain principle, instrumentation and applications of High performance Liquid Chromatography (HPLC).
- ii) Explain principle, instrumentation and applications of Gas Chromatography.
- iii) Explain the principle, instrumentation and applications of IR spectroscopy.

**Q. 3. Answer the following questions (any seven) (7 x 5) = 35**

- i) Distinguish between fluorescence and phosphorescence. Explain factors affecting fluorescence.
- ii) Discuss principle, instrumentation and applications of Gel chromatography.
- iii) Define Chromatography. Classify chromatographic methods with examples.
- iv) Explain the principle of Affinity Chromatography.
- v) Differentiate paper chromatography against TLC with respect to principle and applications.
- vi) Differentiate principle and applications of gel electrophoresis against capillary electrophoresis.
- vii) Differentiate principle and application of nephelometry against turbidimetry.
- viii) Write principle, types and applications of Ion Exchange Chromatography..
- ix) Explain principle, instrumentation and applications of Atomic Absorption spectroscopy.

-----END OF THE PAPER-----

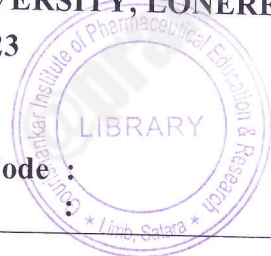
**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**End Semester Examination – Summer 2023**

**Date: 09/06/2023**

**Course : B. Pharmacy**  
**Subject Name : Instrumental Methods of Analysis**  
**Max Marks : 75**

**Sem: VII**  
**Subject Code : BP 701T**  
**Duration : 3 Hr.**



**Instructions:**

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

**Q. 1. Objective Type Questions (Answer all the questions)**

**(10 x 2) = 20**

- i) Name the spectral shifts that occur in UV region.
- ii) Differentiate between isocratic and gradient elution in chromatography.
- iii) Define derivatization. List methods of derivatization in Gas Chromatography
- iv) Mention equations for Beer's and Lambert's Law.
- v) Define quenching. List factors affecting quenching.
- vi) Enlist any four the applications of Atomic Absorption spectroscopy.
- vii) Distinguish between nephelometry and turbidimetry.
- viii) Justify- buffers are used in ion exchange chromatography.
- ix) Define  $R_f$  value. Enlist the factors that affect  $R_f$  value.
- x) How aldehyde and ketone can be differentiated in IR spectrum.

**Q. 2. Long Answers (Answer 2 out of 3)**

**(2 x 10) = 20**

- i) Discuss the principle, instrumentation and applications of HPLC.
- ii) Explain the principle, instrumentation, sampling techniques of IR spectroscopy.
- iii) Describe in brief the principle, instrumentation and applications of gas chromatography.

**Q. 3. Short Answers (Answer 7 out of 9)**

**(7 x 5) = 35**

- i) Discuss the principle of fluorescence using Jablonski diagram.
- ii) Define and classify Ion Exchange resins with examples.
- iii) Describe different development techniques used in Paper Chromatography.
- iv) Explain the principle, theory and applications of gel chromatography.
- v) Explain in brief the various types of electronic transitions occur in UV-Visible spectroscopy.
- vi) Elaborate the different vibrational modes in polyatomic molecules upon IR absorption
- vii) Describe the TLC with different methods for preparations of TLC plates.
- viii) Differentiate between adsorption and partition column chromatography.
- ix) Discuss the principle and interferences occurs in flame photometry

**-----END OF THE PAPER-----**

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2023

Date: 02/01/2024

Course : B. Pharm

Subject Name : Instrumental Method of Analysis

Max Marks : 75

Sem: VII

Subject Code: BP701T

Duration :

3 Hr.

**Instructions:**

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

**Q. 1. Objective Type Questions (Answer all the questions)**

(10 x 2) =

- i) Explain the principle of affinity chromatography.
- ii) Describe principle of gel chromatography.
- iii) Differentiate between normal phase and reversed phase chromatography.
- iv) Which type of columns are used in HPLC ?
- v) Describe ranges of UV-visible radiation?
- vi) Explain role of primary and secondary filters used in photofluorimeter ?
- vii) Define absorbance and transmittance along with their relationship with each other
- viii) "The R<sub>f</sub> value of compound is always less than or equal to one" – Justify statement.
- ix) Explain derivation of Beer-Lambert's law.
- x) Discuss principle of capillary electrophoresis.

**Q. 2. Long Answers (Answer 2 out of 3)**

(2 x 10) =

- i) Describe principle, FT-IR instrumentation and applications of IR spectroscopy.
- ii) Explain principle, instrumentation and applications of Gas Chromatography.
- iii) Describe fluorescence. Discuss factors affecting fluorescence. Explain application of fluorescence spectroscopy.

**Q. 3. Short Answers (Answer 7 out of 9)**

(7 x 5) = 3

- i) Describe different development techniques used in paper chromatography.
- ii) Write a note on different types of transitions in organic molecule.
- iii) Explain the principle, instrumentation of nepheloturbidimetry.
- iv) Explain the principle HPLC? Describe different pumps used in HPLC.
- v) Discuss principle, stationary phases and applications of ion exchange chromatography.
- vi) Explain principle, advantages, disadvantages and applications of TLC.
- vii) Differentiate between FES and AAS with respect to principle, instrumentation and applications.
- viii) What is spectroscopy? Explain different properties of electromagnetic radiation.
- ix) Discuss fundamental modes of vibrations in polyatomic molecule.

-----END OF THE PAPER-----



**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary Summer Examination – 2024**

Course: B. Pharmacy

Semester: VII

Subject Name: Instrumental Methods Of Analysis

Subject Code: BP701T

Max Marks: 75

Date: 12-06-2024

Duration: 3 Hr.

**Instructions to the Students:**

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

|                                 |                                                                                                        |                      |
|---------------------------------|--------------------------------------------------------------------------------------------------------|----------------------|
| <b>Q.1.</b>                     | <b>Objective Type Questions (Answer All the Questions)</b>                                             | <b>(10 X 2) = 20</b> |
| i)                              | Define the terms Accuracy and Precision                                                                |                      |
| ii)                             | Explain in detail Beer's Law with its equation.                                                        |                      |
| iii)                            | Give a sequence of events occurring in Flame Photometry.                                               |                      |
| iv)                             | What different types of Interference and their Corrections in Atomic Absorption Spectroscopy.          |                      |
| v)                              | Write the difference between fluorescence and Phosphorescence.                                         |                      |
| vi)                             | Explain the fundamental vibrations for linear and non linear molecules.                                |                      |
| vii)                            | What is Chromophore and Oxochrome explain with example.                                                |                      |
| viii)                           | Describe Isoocratic and Gradient elution modes in HPLC.                                                |                      |
| ix)                             | Explain the pressed pellet technique in IR.                                                            |                      |
| x)                              | Why water cannot be used as a solvent in IR spectroscopy.                                              |                      |
| <b>Q.2.</b>                     | <b>Long Answers (Answer 2 out of 3)</b>                                                                | <b>(10 X 2) = 20</b> |
| i)                              | Explain in detail theory of IR Spectroscopy and significance of Hooke's law in IR Spectroscopy.        |                      |
| ii)                             | Explain in detail the instrumentation used in Atomic Absorption Spectroscopy                           |                      |
| iii)                            | Give a brief account of Van Demeter Equation.                                                          |                      |
| <b>Q.3.</b>                     | <b>Short Answers (Answer 7 out of 9)</b>                                                               | <b>(5 X 7) = 35</b>  |
| i)                              | Discuss various pumps in HPLC                                                                          |                      |
| ii)                             | Give a brief account of Double beam UV spectrophotometer.                                              |                      |
| iii)                            | What is Bathochromic shift and Hypsochromic shift, give a reason why it occur with a suitable example. |                      |
| iv)                             | Give in detail the principle involved in Turbidimetry.                                                 |                      |
| v)                              | Write a short on modes of vibrations in IR Spectroscopy.                                               |                      |
| vi)                             | Explain in detail the different types of burners used in Flame Photometry                              |                      |
| vii)                            | Give applications of Phosphorescence and fluorescence.                                                 |                      |
| viii)                           | Write a short note on regions in IR spectrum.                                                          |                      |
| ix)                             | Write a short note on Ion Exchange Chromatography.                                                     |                      |
| <b>*** END OF THE PAPER ***</b> |                                                                                                        |                      |

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